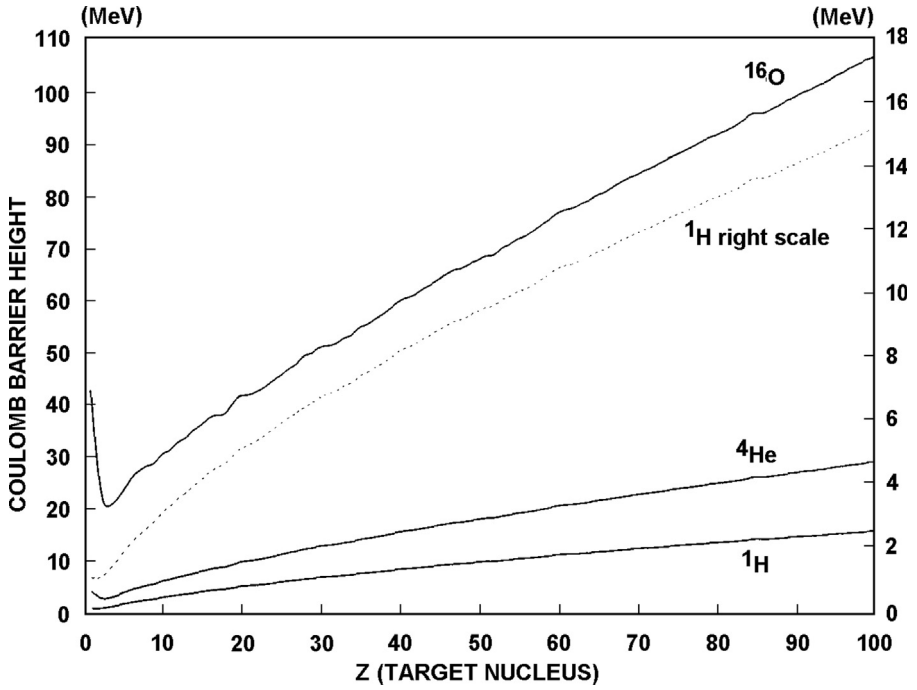


Figure 10.3 The Coulomb barrier height, E_{cb}(min) according to (10.15) for reactions between a target element Z (of most common A) and projectiles ^{1}H, ^{4}He, and ^{16}O.

10.4. RUTHERFORD SCATTERING

If a collimated beam of particles (Z_1, A_1, $E_{\text{kin},1}$) strikes a foil (Z_2, A_2) so that most of the particles pass through the foil without any reduction in energy, it is found that many particles are scattered away from their incident direction (Fig. 10.4). We can neglect multiple scattering and nuclear transformations, since they are many times less than the number of scattering events. Solid angles are measured in steradians, where 1 steradian corresponds to $1/(4\pi)$:th of the surface area of an enclosing sphere such that all particles will cross its surface area.

Geiger and Marsden found that scattering follows the relation

$$\mathrm{d}\sigma/\mathrm{d}\Omega = \left[k\ Z_1\ Z_2\ \mathbf{e}^2/\left\{2m_{\text{red}}\left(v_1^0\right)^2\right\}\right]^2/\sin^4(\theta/2) \tag{10.18}$$

Where $\mathrm{d}\sigma/\mathrm{d}\Omega$ is the differential cross-section (reaction cross-section in barns or m^2 per steradian), θ is the scattering angle (Fig. 10.4) and v_1^0 incident particle velocity. The differential cross-section is a measure of the probability σ of a scattering event per unit solid angle Ω.

$$\mathrm{d}\sigma/\mathrm{d}\Omega = n/(n_{\text{o}}\ N_{\text{v}}\ x\ \Delta\Omega) \tag{10.19}$$

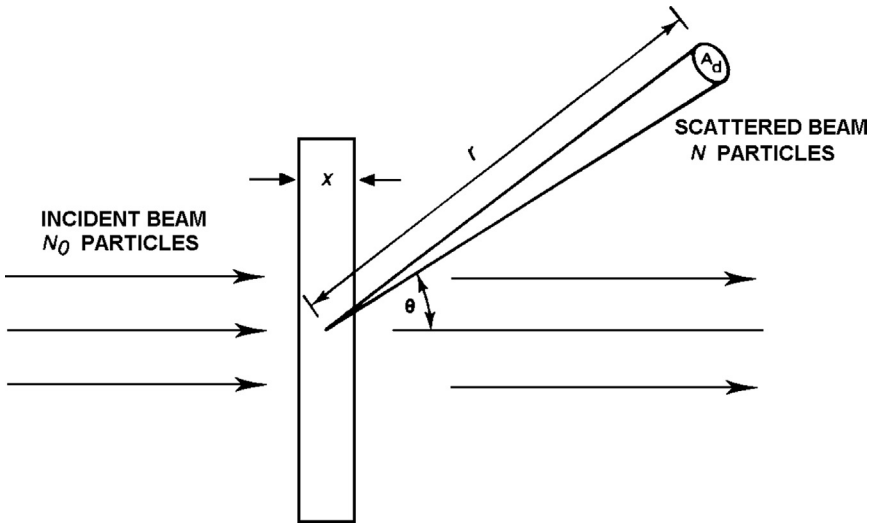


Figure 10.4 Geometry for measuring the scattering by a thin foil.

where n is the number of projectiles scattered into the detector which subtends a solid angle

$$\Delta\Omega = A_{\mathrm{d}}\, r^{-2} \tag{10.20}$$

with respect to the center of the target; A_{d} is the detector area and r is the distance between target and detector; n_{o} is the number of projectiles hitting the target, which is x thick and contains N_{v} scattering atoms per unit volume ($N_{\mathrm{v}} = \rho \mathbf{N}_{\mathrm{A}}/M$, where ρ is the density).

By bombarding foils made of different metals with α-particles from radioactive elements, (10.18) can be used to prove that Z_2 is identical to the atomic number of the metal. This implies that the atom has a central core with a positive charge of that magnitude. Such experiments showed that some particles scattered almost directly back (i.e. $\theta > 90°$), a fact which baffled Rutherford: "It was quite the most incredible event that ever happened to me in my life. It was almost as incredible as if you fired a fifteen inch shell at a piece or tissue paper and it came back and hit you." From the scattering equations he concluded that the scattering center, the "nucleus", had a diameter which was 1000 times smaller than the atom. This same scattering technique was used 55 years later in the US lunar explorer Surveyor 5 (1967) to determine the composition of the lunar surface for the first time.

10.5. ELASTIC SCATTERING

In elastic scattering energy is exchanged between the projectile and the target nucleus but the value of Q is zero:

$$\mathrm{X}_1(v_1) + \mathrm{X}_2(v_2 \approx 0) \;\rightarrow\; \mathrm{X}_1\left(v_1'\right) + \mathrm{X}_2\left(v_2' > 0\right) \quad Q = 0 \tag{10.21}$$